**Abstract**

Blah blah blah

**Definition.** *A completion of a metric space* $(X, d)$ *consists of:*

1. *A complete metric space* $(Z, \rho)$*;*
2. *An isometry* $\iota : X \hookrightarrow Z$ *such that* $\overline{\iota(X)}^{\rho} = Z$.

*We denote the completion of* $(X, d)$ *by the ordered triple* $(Z, \rho, \iota)$.

**Lemma.** *If* $(X, d)$ *is a metric space,* $(Y, \rho)$ *is a complete metric space, and* $\iota : X \to Y$ *is an isometry, then* $(\overline{\iota(X)}^{\rho}, \rho, \iota)$ *is a completion of* $(X, d)$.

**Theorem.** *Every metric space admits a unique completion up to isometric isomorphism.*

*Sketch of Proof.* We will first show uniqueness, then existence.

Let $(Y, \rho, \iota)$ and $(Z, \sigma, \jmath)$ be completions of $(X, d)$.

Let $y \in Y$. Since $\iota(X)$ is dense in $Y$, there exists a sequence $\{\iota(x_n)\}_n \subset \iota(X)$ such that $\iota(x_n) \to y$.

Since the sequence $\{\iota(x_n)\}_n$ is convergent, it is $\rho$-Cauchy.

Since $\rho$ and $\sigma$ are isometries, we have:

$$\sigma(\jmath(x_n), \jmath(x_m)) = d(x_n, x_m) = \rho(\iota(x_n), \iota(x_m)).$$

Hence if $\{\iota(x_n)\}_n$ is $\rho$-Cauchy, then $\{\jmath(x_n)\}_n$ is $\sigma$-Cauchy.

Since $(Z, \sigma, \jmath)$ is complete, $\{\jmath(x_n)\}_n$ is convergent.

Define $\varphi : Y \to Z$ by $\varphi(y) = \lim_{n\to\infty} \jmath(x_n)$. We have the following diagram:

$$\begin{tikzcd} X \arrow[r, "\iota"] \arrow[dr, "\jmath"'] & Y \arrow[d, dotted, "\varphi"] \\ & Z \end{tikzcd}$$

To show uniqueness, show that $\varphi$ is well-defined, an isometry, and bijective.

To show existence: Fix an $x_0 \in X$ and let $x \in X$ be arbitrary. Define $f_x : X \to \mathbb{R}$ by $f_x(t) = d(t, x) - d(t, x_0)$. Show that $f_x \in C_b(X)$ and that $\iota : X \to C_b(X)$ given by $x \mapsto f_x$ is an isometry. ∎

This is called a Kuratowski embedding. Note that $\iota$ is not linear, hence if $X$ was a normed vector space then its image in $C_b(X)$ would not necessarily be a vector space.

**Question.** *Is the completion of a normed vector space also a vector space?*

*Answer.* Yes! Though if we use a Kuratowski embedding on a normed vector space $V$, since $\iota$ is not linear the image of $V$ may not necessarily be a vector space. We need to find a natural linear isometry into a Banach space.

**Definition.** *Let* $(V, \|\cdot\|)$ *and* $(W, \|\cdot\|)$ *be normed vector spaces.*

(1) *The* *operator norm* *of* $T \in \mathrm{Hom}(V, W)$ *is* $\|T\|_{op} := \sup\{\|T(v)\| \mid v \in V, \|v\| \leqslant 1\}$.

(2) $T \in \mathrm{Hom}(V, W)$ *is called* *bounded* *if* $\|T\|_{op} < \infty$. *The set of all bounded linear maps from* $V$ *to* $W$ *is* $\mathbb{B}(V, W) := \{T \in \mathrm{Hom}(V, W) \mid \|T\|_{op} < \infty\}$.

(3) *The* *continuous dual* *of* $V$ *is* $V^* := \mathbb{B}(V, \mathbb{F})$.

*Remarks.*

(1) A linear map is continuous if and only if it is bounded.

(2) $(\mathbb{B}(V, W), \|\cdot\|_{op})$ is a Banach space.

(3) Is $V^*$ nonempty?

**Theorem** (Hahn-Banach)**.** *Let* $X$ *be an* $\mathbb{R}$*-vector space,* $M$ *a subspace of* $X$*, and* $p : X \to \mathbb{R}$ *a seminorm on* $X$*. If* $f : M \to \mathbb{R}$ *is a linear functional on* $M$ *such that* $|f(m)| \leqslant p(m)$ *for all* $m \in M$*, then there exists a linear functional* $F : X \to \mathbb{R}$ *such that* $F|_M = f$ *and* $|F(x)| \leqslant p(x)$ *for all* $x \in X$.

The assumption that $|f(m)| \leqslant p(m)$ implies $f$ is bounded. The claim that $|F(x)| \leqslant p(x)$ implies $F$ is bounded.

**Corollary.** *Let* $(V, \|\cdot\|)$ *be a normed vector space. For any nonzero* $v \in V$*, there exists* $F \in V^*$ *such that* $\|F\|_{op} = 1$ *and* $F(v) = \|v\|$.

*Sketch of Proof.* Fix $v \in V$. Define $M := \{\alpha v \mid \alpha \in \mathbb{R}\}$. This is a subspace of $V$.

Define $f : M \to \mathbb{R}$ by $f(\alpha v) := \alpha \|v\|$. This map is linear.

Note that $|f(\alpha v)| = |\alpha| \|v\| = \|\alpha v\|$. Since every norm is a seminorm, the hypothesis of the Hahn-Banach theorem is satisfied.

Observe that:

$$\begin{aligned}
\|f\|_{op} &= \sup\left\{ |f(\alpha v)| \;\middle|\; \alpha v \in M, \|\alpha v\| \leqslant 1 \right\} \\
&= \sup\left\{ \|\alpha v\| \;\middle|\; \alpha v \in M, \|\alpha v\| \leqslant 1 \right\} \\
&= 1.
\end{aligned}$$

Take $\alpha = 1$, then $f(v) = \|v\|$.

By the Hahn-Banach Theorem, $f$ extends to a linear functional $F \in V^*$. ■

**Corollary.** *Let* $(V, \|\cdot\|)$ *be a normed* $\mathbb{R}$*-vector space and* $v \in V$*. Then* $\|v\| = \sup\{|\varphi(v)| \mid \varphi \in V^*, \|\varphi\|_{op} \leqslant 1\}$.

*Proof.* If $\varphi \in V^*$ and $\|\varphi\| \leqslant 1$, then $|\varphi(v)| \leqslant \|\varphi\|_{op} \|v\| \leqslant \|v\|$. By the previous corollary, □